# NICOLÒ CICCARONE

Senior Data Scientist & Applied Statistician | Machine Learning & MLOps

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## PROFESSIONAL SUMMARY

Applied statistician and senior data scientist with an econometrics background and hands-on experience taking quantitative models into production. Specializations include causal inference, survival analysis, time-series forecasting, climate and financial risk, and machine learning. Builds and optimizes production ML systems using Python, Spark, AWS, model-serving APIs, cloud orchestration and monitoring. Delivered analytics for enterprise clients across staffing, retail, aviation and financial services.

## TECHNICAL SKILLS

- **Statistical Modeling:** Econometrics, causal inference, survival and mixture-cure models, score-driven (GAS) models, forecasting, Hawkes processes, climate-risk scenarios, gradient boosting.
- **Programming & Data:** Python, R, SQL, Polars, PySpark / Apache Spark, Julia (basic), GIS and spatial data, MongoDB.
- **Machine Learning:** PyTorch, mixture-of-experts, NLP document processing, customer lifetime value, churn and demand modeling.
- **MLOps, Cloud & Serving:** AWS SageMaker, ECS, EKS, ECR and CloudWatch; Docker, Kubernetes, FastAPI, Celery task orchestration, Grafana monitoring, production inference and distributed workflows.
- **Additional Platforms:** Databricks; Azure ML, Pipelines and Fabric; GCP Vertex AI Workbench and Colab Enterprise.

## PROFESSIONAL EXPERIENCE

**Complexa Labs** — Bari, Italy
*Quantitative Data Science & Engineering* — 01/2025 – Present

- Deliver statistical modeling and production analytics for enterprise clients with over €1B in revenue.
- Built a time-to-fill forecasting engine using score-driven discrete-time survival models (GAS), right-censored vacancy-day panels and time-varying funnel covariates; deployed in production via AWS SageMaker.
- Developed customer lifetime value models using mixture-cure survival analysis, gradient-boosted margin regression and calibrated segmentation for revenue forecasting and next-best-product targeting.
- Designed and deployed a macroeconomic scenario simulator to forecast business impacts across market conditions, supported by distributed orchestration.
- Refactored production pipelines using Spark-native joins, window functions and broadcast optimization, **reducing execution time by 70%**; delivered NLP document-processing algorithms as FastAPI microservices.

**acmetric B.V.** — Amsterdam, Netherlands
*Data Scientist* — 01/2024 – 11/2024

- Developed and deployed causal AI pricing and demand-forecasting models for Adidas International B.V.; the pricing algorithm remains in production use.
- Developed and maintained churn / litigation prediction models for LATAM Airlines using gradient boosting.

**MOOS Solutions B.V.** Amsterdam, Netherlands
*Data Scientist* 10/2022 – 12/2023

- Designed and deployed a proprietary mixture-of-experts engine for real-time inventory optimization, routing spatial IoT sensor streams to specialist sub-models; used by a startup, with reported accuracy of up to **90%** (metric to be specified).

**UBS** Kraków, Poland
*Risk Modeling and Analytics Specialist* 04/2022 – 09/2022

- Developed and deployed a first climate-risk model for a commercial real-estate portfolio, applying scenario-based stress tests aligned with NGFS / IPCC pathways.

## EDUCATION & SPECIALIST TRAINING

**Università Cattolica del Sacro Cuore** 04/2026 – 07/2026
*Specialized Training Programme in Statistical Epidemiology*
Methods for causal attribution of environmental exposures to cardiovascular, respiratory and immune-mediated public-health outcomes.

**Tinbergen Institute, Amsterdam** 07/2023
*PhD-Level Summer School in Deep Learning and Generative AI*
Intensive training with Eran Raviv, PhD.

**Vrije Universiteit Amsterdam** 2021 – 2022
*M.Sc. Econometrics & Operations Research*
Grade: 8.0/10. Thesis: *Price volatility and climatic extremes: distinguishing signal from noise.*

**University of Bologna** 2019 – 2021
*M.Sc. Economics, Econometrics Track*
Grade: 110/110 cum laude. Thesis: *Hedging against climate change: A Hawkes process approach.*

**University of Trento** 2015 – 2018
*B.Sc. Economics and Management*
Grade: 110/110 cum laude. Department of Economics Honours Programme.

## LANGUAGES & RECOGNITION

- **Languages:** Italian (native), English (C1/C2), French (B1), German (A2).
- **Recognition:** INPS Merit Scholarship (2019); University of Trento Merit Scholarship (2018).
- **Certification:** TOEFL 107/120 (2021).